GO-BACK-N PROTOCOL

Aim: To implement the Go-Back-N ARQ protocol using C programming.

Program

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int i,n,ws;

    clrscr();

    printf("Enter the number of frames: ");
    scanf("%d",&n);

    printf("Enter the window size: ");
    scanf("%d",&ws);

    printf("\nGo Back N Protocol\n");

    for(i=1;i<=n;i++)
    {
        if(i<=ws)
        {
            printf("\nFrame %d sent",i);
            printf("\nFrame %d received",i);
        }
        else
        {
            printf("\nWindow slides");

            printf("\nFrame %d sent",i);
            printf("\nFrame %d received",i);
        }
    }

    printf("\n\nAll frames transmitted successfully");

    getch();
}
```

Sample Output

```
Enter the number of frames: 5
Enter the window size: 3

Go Back N Protocol

Frame 1 sent
Frame 1 received
Frame 2 sent
Frame 2 received
Frame 3 sent
Frame 3 received
Window slides
Frame 4 sent
Frame 4 received
Window slides
Frame 5 sent
Frame 5 received
All frames transmitted successfully
```

Algorithm

1. Start the program.
2. Read the number of frames.
3. Read the window size.
4. Send frames according to the window size.
5. Receive acknowledgement for each frame.
6. Slide the window and send the next frames.
7. Display successful transmission.
8. Stop the program.

Result

Thus, the Go-Back-N protocol program was implemented successfully using Turbo C.

Note: This is a simple demonstration of window movement. It does not simulate frame loss, retransmission, or complete Go-Back-N error recovery.